Entrepreneurial Project: **Mirae Daw — Look Into Your Light**

1. Project Overview

Mirae Daw is a digital wellness and journaling experience designed to help young adults rediscover emotional clarity through reflection, guided prompts, and mindful interaction.

The name **Mirae Daw** blends two powerful ideas: *"Mirae,"* the Korean word for *future* — a symbol of hope and what is yet to come — and *"Daw,"* derived from the Arabic word for *light.* Together, they represent *the light of the future* — a reminder that healing, growth, and self-understanding are part of an ever-evolving journey.

Acting as an empathetic digital mirror, **Mirae Daw** responds to your emotions with personalized journaling guidance, mood-based design, and an AI companion that nurtures self-awareness and emotional growth.

Rather than replacing therapy, **Mirae Daw** serves as a gentle bridge between professional support and personal reflection — a safe, creative space where users can explore, express, and illuminate their path forward with hope and compassion.

2. Stakeholders

Key stakeholders include:

- - Primary Users (Teens & Young Adults): Daily users of the app – for emotional relief and self-reflection
- - Parents/Caregivers: Indirect beneficiaries – may see improved emotional awareness in children
- - Schools/Universities: Potential partners – could use it as a student support tool
- - Mental Health Advocates: Influencers – may endorse or promote wellness tech
- - UX Designers: Co-creators – contribute to the emotional aesthetic and UI
- - Investors/Startups: Future scalability – business growth and monetization options

3. Stakeholder Priority Matrix

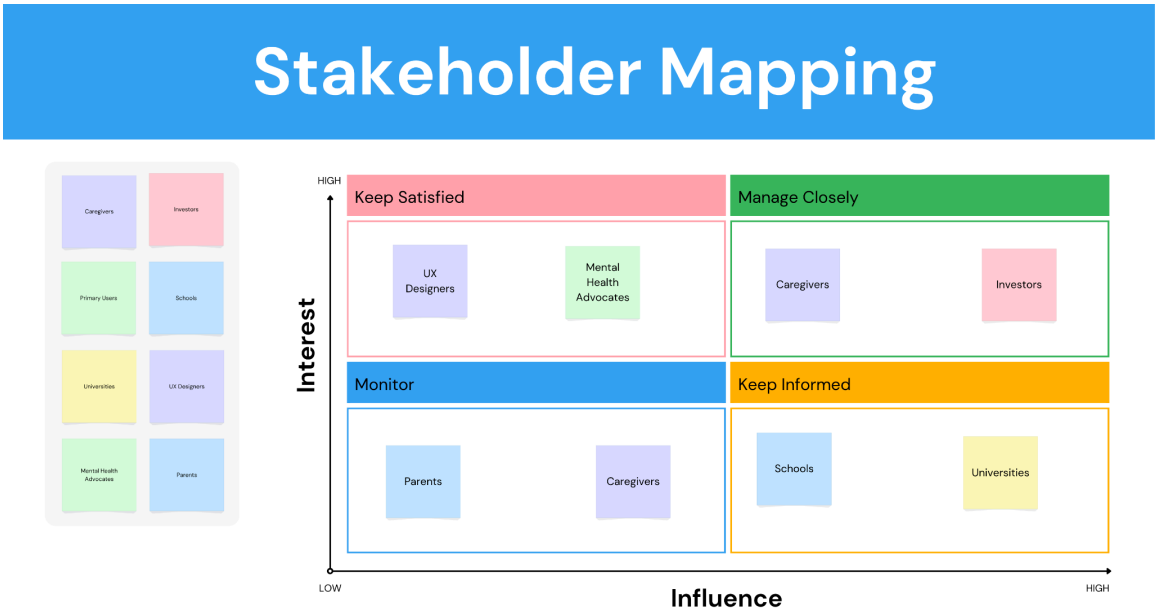
Stakeholder mapping based on their influence and interest:

High Influence / High Interest: Investors, Primary Users

High Influence / Low Interest: Schools/Universities

Low Influence / High Interest: UX Designers, Mental Health Advocates

Low Influence / Low Interest: Parents/Caregivers



4. Proposed Tech Stack

Technology	Purpose	Experience
Springboot	Backend API	🟢 Familiar
Native wind (local to react native)	Styling with utility classes	🟢 Familiar
Framer Motion	Page animations & transitions	🟢 Familiar
Firebase	Authentication, Firestore DB, Hosting	🟢 Familiar
Google Cloud Vision API	Mood detection via images (optional)	🔴 New
LangChain / OpenAI API	AI journaling assistant prompts	🔴 New

Recharts or D3.js	Mood tracking visualizations	🔴 New
Electron.js	Cross-platform desktop version	🟢 Currently learning
React native	Mobile Application	
Vercel	Hosting Next.js frontend	🟢 Familiar
Figma	UI/UX prototyping	🟢 Familiar
Ready Player Me	3d avatar	
Live link (Unreal engine)	3d avatar animations (facial expressions)	
Control Rig, IK retargeter, Motion warping and animation blueprint (Unreal engine)	3d avatar animations (body animations)	

5. Development Timeline (Initial Draft)

Phase 1: Ideation & Research (Sept - Oct)

- Stakeholder interviews and occasional supervisor meetings
- Define user personas and mental health/student support needs
- Initial moodboard and concept sketches

Phase 2: AI Agent Development (Oct - Jan)

- Develop the core AI agent (main workload)
- Implement mood analysis, journaling interactions, and response logic
- Iterative testing and refinement of AI capabilities

Phase 3: Mobile App Integration (React Native) (Jan - Feb)

- Integrate AI agent into the React Native mobile app
- Implement user authentication (Firebase Auth), mood selector, and journaling system
- Feedback loop with supervisor and initial testing

Phase 4: Optional Desktop Version (Electron.js) (Feb - March)

- Develop Electron.js version of the app (if feasible)
- Ensure AI agent works seamlessly across both platforms

Phase 5: Reporting & Documentation (March - April)

- Prepare project report and documentation
- Record AI agent performance, user interactions, and development notes

Phase 6: Stakeholder Presentation & Project Wrap-Up (April)

- Meet with mental health/student support department to present the tool
- Gather feedback and discuss potential implementation
- Final supervisor meetings and project refinements

Phase 7: Final Deployment & Submission (April - May)

- Final app deployment on mobile (and desktop if applicable)
- Submit project as final year work

6. Risks & Mitigations

Risk	Likelihood	Mitigation
Over-scoping AI features	Medium	Focus on GPT journaling & sentiment only

Time management	High	Agile sprints, weekly milestones
Access to user testers	Medium	Survey students or online feedback forms
Unfamiliar tech	Medium	Schedule weekly learning hours

7. Suitability for CO3202

This project demonstrates a strong entrepreneurial opportunity by merging technical challenges with real-world emotional wellness needs. It fits CO3202 as it addresses a social issue (mental health), targets a clear user base, and introduces scalable, creative tech. The project also encourages empathy, adaptability, communication, and long-term leadership skills.

8🏗️ The Pipeline (Your AI Agent Architecture)

Let's map your **big agent** into **sub-agents / tools**. Each one has a clear job.

1. Input Stage

- **Journal Entry** (text)
- **3D Model (GLB)**

---

2. Text Analysis Sub-Agent

- **What it does**: Takes the journal entry → outputs emotion/vibe.
- **Tools**:
    - Hugging Face (`transformers`) for sentiment/emotion classification.
    - Your **database** to fine-tune (train a custom vibe model).

Output:

```
{
  "emotion": "anxiety",
  "vibe_phrase": "feeling overwhelmed but hopeful"
}
```

---

3. Animation Mapping Sub-Agent

- **What it does**: Translates emotion/vibe into animation instructions.
- **Tools**:
  - A **lookup table (dictionary/JSON)** mapping emotions → animation actions.

Example:

```
EMOTION_TO_ANIMATION = {
  "happy": {"face": "smile", "posture": "upright"},
  "sad": {"face": "frown", "posture": "slouch"},
  "anxious": {"face": "tight lips", "gesture": "fidget"}
}
```

  -

Output:

```
{
  "face": "frown",
  "posture": "slouch"
}
```

---

4. Avatar Animation Sub-Agent

- **What it does**: Takes animation instructions → applies them to the GLB model.
- **Tools**:
  - **AvatarSim** (Python-controlled → Unreal Engine).
  - **UE Control Rig & LiveLink** for real-time body/facial animation.

Output:

- A **live animated avatar** in Unreal (for interactive apps), or
- An **exported animation file** (FBX/GLB/MP4).

---

5. Output Stage

- Final deliverable = Animated 3D avatar that reflects the journal entry's mood.

🔧 Tools You'll Need

- **NLP** → Hugging Face (`transformers`, sentiment/emotion model).
- **Animation logic** → A simple **Python dictionary** mapping vibes to actions.
- **Animation executor** → AvatarSim + UE LiveLink + Control Rig.
- **Optional**: Text-to-speech (e.g., `pyttsx3`, ElevenLabs, or Hugging Face TTS models) if you want the avatar to *speak* the vibe phrase.